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ORIGINAL RESEARCH

TSD-YOLO: Small traffic sign detection based on improved YOLO v8

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Abstract

Traffic sign detection is critical for autonomous driving technology. However, accurately detecting traffic signs in complex traffic environments remains challenge despite the widespread use of one-stage detection algorithms known for their real-time processing capabilities. In this paper, the authors propose a traffic sign detection method based on YOLO v8. Specifically, this study introduces the Space-to-Depth (SPD) module to address missed detections caused by multi-scale variations of traffic signs in traffic scenes. The SPD module compresses spatial information into depth channels, expanding the receptive field and enhancing the detection capabilities for objects of varying sizes. Furthermore, to address missed detections caused by complex backgrounds such as trees, this paper employs the Select Kernel attention mechanism. This mechanism enables the model to dynamically adjust its focus and more effectively concentrate on key features. Additionally, considering the uneven distribution of training data, the authors adopted the WIoUv3 loss function, which optimizes loss calculation through a weighted approach, thereby improving the model's detection performance across various sizes and frequencies of instances. The proposed methods were validated on the CCTSDB and TT100K datasets. Experimental results demonstrate that the authors' method achieves substantial improvements of 3.2% and 5.1% on the mAP50 metric compared to YOLOv8s, while maintaining high detection speed, significantly enhancing the overall performance of the detection system. The code for this paper is located at https://github.com/dusongjie/TSD-YOLO-Small-Traffic-Sign-Detection-Based-on-Improved-YOLO-v8

1 | INTRODUCTION

Traffic signs play a crucial role on highways and urban roads, ensuring road traffic safety, and enhancing traffic efficiency. Fast and precise detection and recognition of traffic signs are crucial aspects of environmental perception in autonomous driving. It also represents a prominent area of current research.

Deep learning and computer vision technology have driven research direction in traffic sign detection towards deep convolutional neural network. In autonomous driving scenarios, intricate challenges arise, including the small size of traffic signs and instances of occlusion. Developing a traffic sign detection algorithm that balances lightweight efficiency and robust detection performance is an urgent and critical challenge.

Currently, object detection algorithms are categorized into two primary groups: one-stage detection algorithms and two-stage detection algorithms. This categorization is based on whether candidate boxes are generated through independent learning or direct sampling of dense regions. One-stage object detection algorithm excels in detection speed compared to two-stage detection algorithm, but they often exhibit lower accuracy. However, in recent years, we have seen the emergence of numerous training strategies tailored for convolutional neural networks. These strategies have substantially increased the detection accuracy of the one-stage detection algorithms. Consequently, one -stage detection methods have become prominent as the preferred approach for traffic sign detection.

Since Redmon first introduced the YOLO [1] (You Only Look Once) detection algorithm, it has undergone rapid development in recent years, owing to its notable advantages in both high detection accuracy and speed. This evolution has resulted in the creation of various versions [2–6]. Currently, YOLOv8 is widely utilized due to its accurate and rapid detection performance. YOLOv8 is categorized into five models of varying sizes, determined by scaling coefficients. Among these, YOLOv8s emerges as a smaller version with fewer network layers and parameters. Although it is inferior to the large model in detection accuracy, it boasts faster detection speed. However, in complex traffic scenarios, the multi-scale variations of traffic signs, interference from background elements, and imbalanced sample capture can cause model misdetections and inconsistent recognition capabilities. To address these issues, this paper proposes an enhanced traffic sign detection method based on the YOLOv8s model. The specific contributions include the following:

(i) This paper applies the Select Kernel (SK) attention mechanism within the neck network. This mechanism dynamically adjusts the weights of the feature channels to enhance the focus and recognition of critical features. It is particularly effective in addressing challenges posed by complex backgrounds and occlusions commonly found in traffic scenes, which can result in missed detections.

(ii) The Space-to-Depth (SPD) module is integrated into the backbone network to distribute spatial information into depth channels. This adjustment accelerates the feature extraction process and expands the model's receptive field. This effectively improves the model's capability to detect traffic signs across various scales, thereby reducing instances of missed detections.

(iii) The WIoUv3 loss function is employed as the bounding box loss function in this study. Assigning different weights to each sample significantly enhances the model's ability to handle objects of various sizes and frequencies. This approach is particularly effective for detecting small or rare traffic signs, ensuring accurate recognition of these crucial elements.

2 | RELATED WORK

2.1 | Traffic sign detection

Traffic sign detection is a prominent research direction in the field of autonomous driving. Numerous scholars have proposed a wide array of algorithms for classifying and detecting road traffic signs. These algorithms can be roughly categorized into two groups, the first relies on traditional detection techniques, including methods based on colour, shape features, and feature fusion; the second group is founded on deep learning. Due to the imperative of real-time performance in the context of autonomous driving, the current traffic sign detection methods are predominantly based on one-stage detection algorithms.

One-stage detection algorithms, such as Single Shot MultiBox Detector (SSD) [7], YOLO, and RetinaNet [8], have the capability output both the location and category of the bounding box based on the features extracted by the backbone network. Among these, the YOLO series stands out as a classic single-stage detection algorithm. It excels by predicting the object's position and categories in a single pass through the network, defining the object detection task as an end-to-end regression problem. Following YOLO, SSD, and RetinaNet introduced significant enhancement. The SSD algorithm fed various levels of features into the detection module, thereby improving the accuracy of small objects. Additionally, SSD borrowed the concept of prior boxes and set different scales for each grid, reducing the training complexity. RetinaNet introduced the Focal Loss function, which dynamically adjusts the cross-entropy loss function based on the confidence level. This innovative enables the model to tackle the issue of imbalanced positive and negative samples, resulting in a further improvement in the accuracy of the one-stage detection algorithms.

Currently, within the domain of autonomous driving, numerous traffic sign detection algorithms based on YOLO have emerged. Yu [9] observes that traditional traffic sign detection typically involves individual image-based detection and recognition, overlooking valuable information within image sequences. To address this, they introduced a fusion model that combines YOLOv3 and VGG19, leveraging the sequential relationships between multiple images to enhance the accuracy of traffic sign detection. Different layers feature maps contain varying object-related information, with richer information contributing to improved detection and classification performance. Zhang [10] introduced a spatial information aggregator to integrate the spatial information from lower-level feature maps into higher-level ones. They also proposed a multi-scale feature attention module to refine the fused feature information and enhance the feature pyramid. Yao [11] refined the feature fusion approach from YOLOV4 by introducing an adaptive feature pyramid network (AFPN) to dynamically fuse features of different scales. Simultaneously, a receptive field module composed of multi-branch expanded convolution was added to the backbone network to improve the feature extraction capability of the detector. Liang proposed a one-stage detection algorithm fortified by transformer structure [12]. This approach facilitates the acquisition of object category relationships and enhances model sensitivity through implicitly learning of inter-object relationships. Additionally, a Global Extraction Encoder (GEE) structure was designed to enhance the global perception ability. Khokhor [13] utilized the YOLOv3 model to design an Automatic License Plate Recognition (ALPR) system, achieving lightweight and efficient license plate detection. Chen et al. [14] employed semantic priors and deep attention residual groups to adaptively fuse high-level semantic information with low-level texture information, effectively validating the importance of different hierarchical features, which enlightens the feature extraction process in object detection. Wang [15] has noted that the scale of traffic signs varies significantly, leading to fluctuations in detection accuracy. Furthermore, they have observed that the original feature pyramid network may not adeptly extract multi-scale feature maps, thus compromising the feature consistency across different traffic signs. To address these issues, they present an improved feature pyramid model, incorporating an adaptive attention module and a feature enhancement module. This innovation reduces information loss during multi-scale feature fusion, enhances the feature representation ability of the feature pyramid, preserves more details, and improves model's capability to adapt to the diversity of traffic signs characteristics. The complexity of the traffic environment presents a challenge in striking a delicate balance between real-time performance and accuracy. In this context, Chu [16] introduced a lightweight multi-branch detection head based on YOLOv5. This detection head adopted the prevailing decoupling head approach to suppress redundant features, coupled with the incorporation of a global feature extraction module. These enhancements not only elevate the feature extraction process but also bolster the internal correlation of features. As a result, detection accuracy experiences a noticeable boost while maintaining real-time capabilities. Chen [17] proposed a partial multi-scale channel attention mechanism to address the multi-scale processing issue, enhancing the model's focus on low-level texture features and utilizing low-level feature information for extracting more complete and stronger high-level semantic features. Chen [18] and others highlighted the issue of information loss during the feature extraction process, particularly regarding low-level texture information, which is crucial for the overall model representation. Therefore, they proposed a U-Net-like network that compresses channels while extracting multi-level features. This approach fuses compressed features at the bottom and compensates during the upsampling process, effectively mitigating feature information loss. Kedia [19] have significantly enhanced license plate detection capabilities in challenging environments by optimizing the YOLO framework, particularly improving detection accuracy under conditions of insufficient lighting. Wei et al. [20] designed a single-stage detector called YOLOF-F, introducing an Feature Fusion Module (FFM) module to fuse features of different scales and an angle encoder, Computer Vision and Image Understanding (CAVE), to enhance the information of angle points information in the feature maps, thereby improving localization accuracy. Zhang et al. [21], to cope with traffic sign detection under complex weather conditions, designed an image adaptive module to enhance the model's robustness. In the detection network, they designed a multi-layer perceptron module and a feature aggregation module to effectively fuse features of different resolutions; they also utilized a receptive field expansion module to generate output features with different receptive fields to compensate for the deficiencies of singular features. Hu et al. [22] utilized MobileNetv3 as the backbone network to extract multi-level features. They employed an attention mechanism in the neck network to optimize the fusion phase of effective features, ultimately achieving high detection speed and accuracy.

The two-stage object detection network introduces RPN and other regional generation networks, resulting in slower detection speed but heightened detection accuracy. Notable examples of two-stage detection algorithms include Region-based Convolutional Neural Network (R-CNN) [23], FAST R-CNN [24], and Faster R-CNN [25], all of which have been widely employed. Several existing traffic sign detection methods have drawn upon enhancements inspired by these two-stage detection network algorithms. Liang [26] proposed an improved sparse R-CNN, integrating the coordinate attention module with ResNeSt to make the extracted features focus more on important information. Simultaneously, they introduce adaptive enhancement and detection time enhancement modules to elevate both the detection accuracy and algorithmic robustness. Li implemented automatic data augmentation technology in Faster R-CNN and incorporated an attention mechanism to guide the feature pyramid network [27], mitigating the loss of context information and improving the accuracy. However, despite these improvements, the detection speed remains insufficient to meet the real-time requirements. In two-stage-based traffic sign detection, candidate boxes are generated using a candidate box generator like RPN, followed by the execution of classification and regression on these candidate boxes. Li et al. [28] designed a simple yet effective two-stage fusion network that directly predicts categories in the first stage of detection and fuses them with the categories predicted in the second stage to enhance overall robustness. In addition, they developed a post-processing technique: Surround Awareness Non-Maximum Suppression (SA-NMS), which more effectively filters detection boxes. However, when dealing with small objects such as traffic signs, the candidate box generator may produce a reduced number of candidate boxes, or the generated candidate boxes may exhibit lower quality, failing to accurately encompass the small object. This may lead to missed detection or false detection of small objects. Two-stage detection algorithms typically require an extensive number of candidate box generation and classification operations. This can result in significant consumption of consumption of computational and memory resources, making it challenging to meet the real-time requirements in autonomous driving scenarios.

The detection speed of one-stage detection network aligns well with the demands of real-time traffic sign detection. Furthermore, an increasing number of methods and training strategies are emerging to improve the accuracy of single-stage detection algorithms. However, when dealing with small objects like traffic signs, the issue of information loss inevitably arises during the convolution or downsampling processes, presenting a bottleneck that affects the detection performance. The multi-scale variations in traffic signs further compound the challenges in the detection process. Utilizing an appropriate receptive field for feature extraction is essential for making objects of varying scales better suited for the dynamic traffic environment. Incorporation of an attention mechanism and feature enhancement module to preserve more feature information will inevitably increase the number of model parameters, consequently diminishing the detection speed. Therefore, enhancing detection accuracy while maintaining a balanced detection speed remains a challenge to be addressed.

2.2 | Version evolution of YOLO

The YOLO series represents the most up-to-date algorithm in the single-stage detection network, and its performance undergoes continuous optimization to meet the demands of increasingly intricate scenarios. Compared with traditional object detection algorithms, YOLOv1 excels at achieving real-time object detection in real-world scenes. However, YOLOv1 faces issues such as reduced localization precision, constrained classification capability, and inconsistent handling of objects across various scales. To address the aforementioned challenges, YOLOv2 introduces the concept of anchor boxes, enhancing object localization precision. Additionally, it employs Darknet-19 to extract features across various scales, thereby enhancing the model's ability to detect objects of different sizes. However, its training process presents a notable level of complexity, and it encounters the challenge of imbalance between the positive and negative samples. Darknet-53 is employed as the backbone network in YOLOv3, and the multi-scale prediction layers are introduced to enable object detection across diverse feature maps. Meanwhile, feature fusion is executed on the feature maps of varying scales to enhance the overall performance of the detector. Due to the utilization of a more intricate backbone network, its speed is constrained, and its capacity to detect small objects is inadequate. YOLOV4 incorporates Spatial Pyramid Pooling (SPP) and Path Aggregation Network (PANet) to enhance the model's feature extraction and perception capabilities. Additionally, it employs a data enhancement strategy to improve the model's generalization capacity. However, it demands more substantial computing resources and still exhibits limitations in detecting small objects. YOLOv5 optimizes the network structure, reducing model parameters and calculation load. It introduces additional training techniques and strategies, accelerating model convergence, and training efficiency. This results in improved regression and classification capabilities for the object box, ultimately enhancing detection accuracy. YOLOv7 uses an accelerated convolution operation to achieve superior detection speed, and integrates the ELAN strategy to enhance the deep model's learning and convergence. It places substantial demands on computational resources and could experience an elevated false detection rate due to the utilization of a large receptive field. The C2f module proposed by YOLOv8 enhances gradient flow within the network, resulting in an accelerated NMS process. Simultaneously, the utilization of the decoupling head refines the training strategy, elevating its overall detection performance. Nevertheless, challenges persist in the field of small object detection.

3 | PROPOSED METHOD

In the context of autonomous driving, traffic signs detection continues to face several challenges. For instance, during real-time driving, the size of the traffic sign can dynamically change as the vehicle moves. To ensure that the vehicle has adequate time to address complex traffic situations, the detection algorithm must accurately identify traffic sign when they appear at a small size.

To address the aforementioned challenges, this paper propose a method based on YOLOv8s. Specifically, it incorporates a novel convolution module known as SPD [30], designed to handle low-resolution images and small objects. Furthermore, it replaces pooling and stride volume operations with convolution, mitigating the loss of fine-grained information during the feature extraction process. When dealing with small traffic signs, the initial large object detection head is removed. Instead, there is a fusion of shallow features from the P3 level with deep features. Subsequently, a Select Kernel (SK) [29] attention module is utilized to dynamically adapt the receptive field for small object detection, enriching the feature information, and making it more comprehensive. Finally, the feature map is fed into the decoupling head, which serves as the minimal object detection head. During the prediction stage, the WIoUv3 [31] loss is employed as the bounding box loss function. A better gradient gain allocation strategy is proposed inspired by the dynamic non-monotonic focusing mechanism, which effectively improves the network's regression accuracy. The improved network structure is illustrated in Figure 1. As depicted in Figure 1, this paper incorporates the SPD module into the backbone network of YOLOv8 during the initial feature extraction phase. This module compresses spatial information into the channel dimension, preserving finer details, mitigating information loss from stride convolution in the downsampling, accelerating feature extraction, and enlarging the receptive field. This improves the detection capabilities for traffic signs of varying scales, addressing missed detections due to multi-scale object transformations. In the neck network, the SK attention module is utilized to further fuse features, enabling the model to prioritize object features over complex background elements, thus alleviating detection errors. Finally, the WIoUv3 loss function is employed in the detection head as the bounding box loss function. This function assigns weights to samples based on their occurrence frequencies, enhancing the model's ability to detect challenging samples, improving generalizability and, mitigating issues related to missed and false detection.

3.1 | SK module

Traditional CNNs often employ fixed-size convolutional kernels for features extraction in traffic sign detection, leading to potential neglect or blending of feature information across different scales. Consequently, the network may fail to fully leverage feature information across various scales, thereby limiting the detector's performance. Hence, this paper introduces the SK attention module, enabling the network to selectively adjust the importance of features across diverse scales. By introducing branch structures with convolutional kernels of varying sizes, the SK attention mechanism can capture feature information across multiple scales. This enhances the network's perception of features at diverse scales and improves its adaptation to objects of varying scales. Additionally, the SK attention module selects features from different branches by learning adaptive

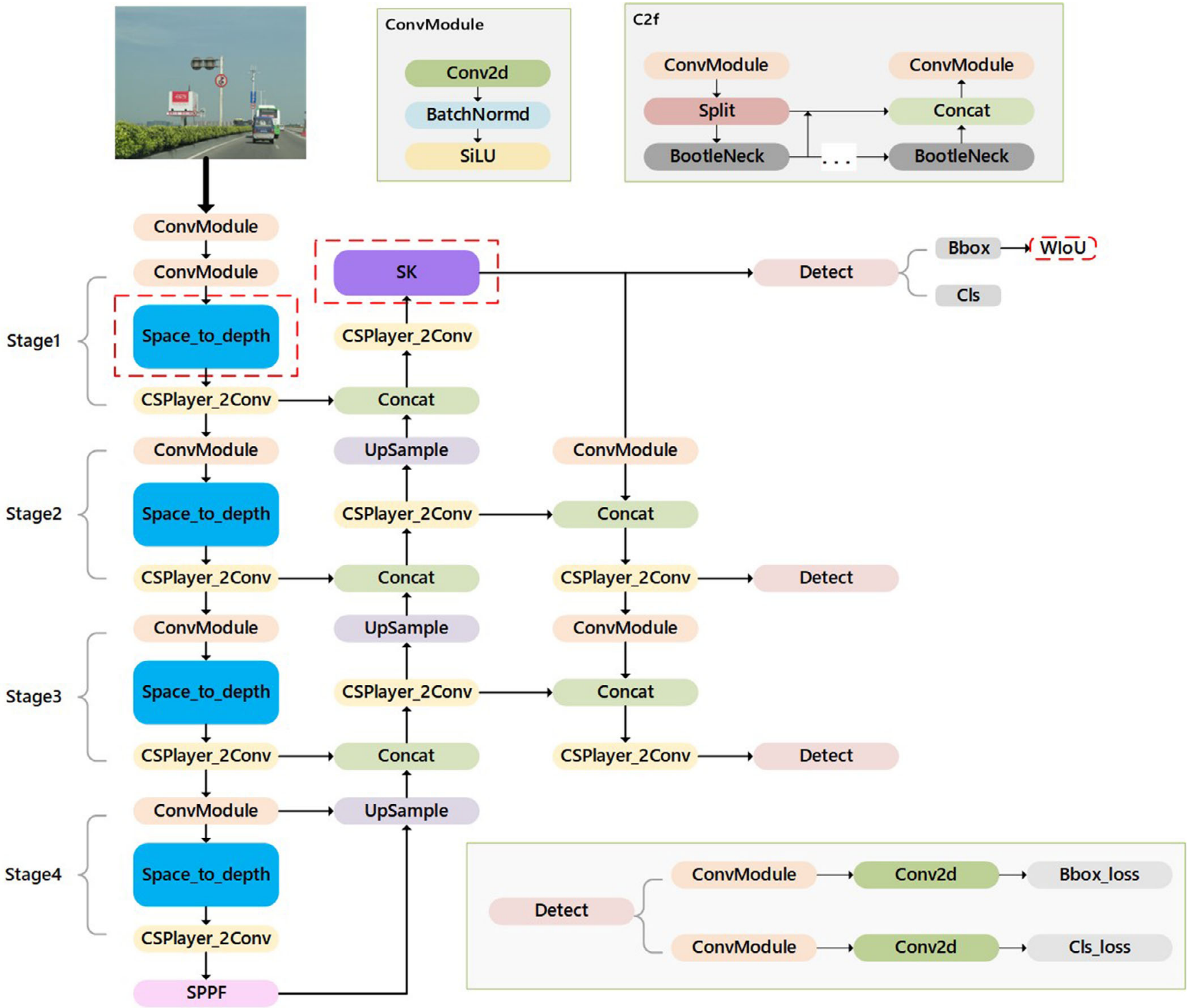

FIGURE 1 Traffic Sign Detection (TSD)-YOLO structure. YOLO, you only look once.

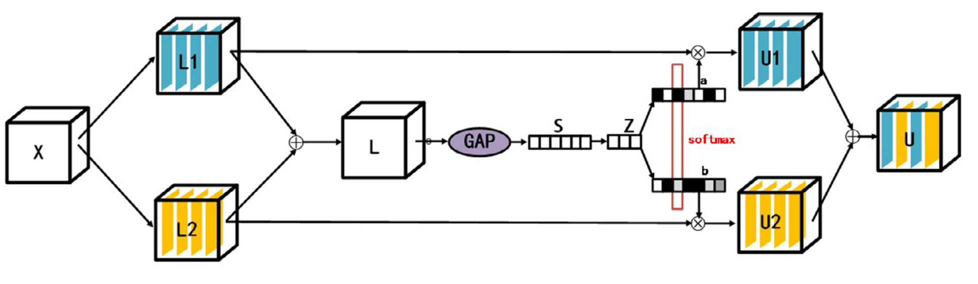

FIGURE 2 SK module. SK, Selective Kernel.

weights. These weights indicate the contribution of each branch to the final feature representation. By dynamically adjusting these weights, the SK attention mechanism automatically selects the most representative features according to the specific task and input data requirements, thereby enhancing the detector's ability to handle multi-scale variations in traffic sign detection. The structure of the SK module is illustrated in Figure 2.

The SK attention module comprises three stages: split, fuse, and select. The subsequent steps delineate a dual-branch configuration as an example:

3.1.1 | Split stage

The input feature map, denoted as X, is fed into two branches with convolutional kernels of sizes 3×3 and 5×5, respectively. Each branch undergoes convolution, normalization, and Rectified Linear Unit (ReLU) activation, resulting in intermediate feature maps denoted as $L1$ and $L2$. $L1$ and $L2$ represent feature maps extracted with different receptive fields.

3.1.2 | Fuse stage

The feature maps generated from the two branches, $L1$ and $L2$, are combined via element-wise summation to create a multi-scale feature map denoted as $L = L1 + L2$. Then, global average pooling is applied to L to reduce its dimensionality, resulting in a feature vector denoted as S. This feature vector S is then fed into a fully connected layer followed by an activation function, producing a vector denoted as Z. This vector Z contains attention weights corresponding to each branch.

3.1.3 | Select stage

The feature vector Z undergoes processing by two functions, A_c and B_c. The values obtained from these functions are element-wise multiplied with $L1$ and $L2$, resulting in feature vectors $U1$ and $U2$, respectively. Finally, $U1$ and $U2$ are element-wise summed to obtain the final feature map, denoted as U.

The pseudocode description of this process is as follows: **Process 1**: Select Kernel

```
function Selcect Kernal Attention (input_feature_map X);
Input: featuremap X
Output:featuremap U
# Split Stage
L1 = Convolution(X, kernel_size = 3 × 3)
L1 = Normalization(L1)
L1 = ReLU(L1)
L2 = Convolution(X, kernel_size = 5 × 5)
L2 = Normalization(L2)
L2 = ReLU(L2)
# Fuse Stage
L = L1 + L2
S = Global_Average_Pooling(L)
Z = Fully_Connected_Layer(S)
Z = Sigmoid(Z)
# Select Stage
U1 = Elementwise_Multiply(Z, L1)
U2 = Elementwise_Multiply(Z, L2)
U = U1 + U2
return U
```

Attention weights for each branch are calculated using a small multi-layer perceptron after global pooling. These weights indicate the contribution of each branch to the final feature representation. Thus, by considering the magnitudes of the attention weights, the module can dynamically select and adjust features from various branches.

3.2 | SPD module

Stride convolution or pooling operations may result in the loss of fine-grained information and inefficient feature representation learning, leading to decreased performance in detecting low-resolution and small objects. SPD-Conv, a novel Convolutional Neural Network (CNN) module, can replace stride convolution or pooling operations for downsampling. This reduces the loss of fine-grained details and enhances adaptability to tasks like detecting small traffic sign objects.

The SPD-Conv module comprises two parts: the SPD layer and a non-stride convolution layer. Within the SPD layer, the feature map undergoes slicing based on a scaling factor to achieve downsampling. For an input feature map of size $S \times S \times C$, the slicing operation is based on the scaling factor, resulting in $scale \times scale$ feature submaps. Each submap has a size of $(S/scale) \times (S/scale) \times C$. These submaps are concatenated along the channel dimension to yield a feature map X' of size $(S/scale) \times (S/scale) \times C$, where $C1 = C \times scale^2$. In essence, the spatial dimension of the feature map X decreases by a scaling factor, while the channel dimension increases by the same factor.

The pseudocode description of this process is as follows:

Process 2: SPD

```
function Space-to-Depth(input_featuremap, scale);
Input: input_featuremap
Output: output_featuremap
S = input_feature_map.height
C = input_feature_map.channel
Initialize output_feature_map with dimensions [(S/scale), (S/scale), C * (scale^2)]
for i from 0 to scale - 1:
  for j from 0 to scale - 1:
    submap = input_feature_map.slice(start_row = i*(S/scale), end_row = (i+1)*(S/scale), start = j*(S/scale), end = (j+1)*(S/scale))
    output_feature_map.append_channels(submap)
output_feature_map.reshape([S/scale, S/scale, C * (scale^2)])
return output_feature_map
```

Next, a non-stride convolution is applied to alter the feature map dimension. Non-stride convolution is chosen for its ability to preserve as much feature information as possible. Similar to stride convolution or pooling operations, SPD-Conv 'shrinks' the feature map. However, SPD-Conv maximizes the utilization

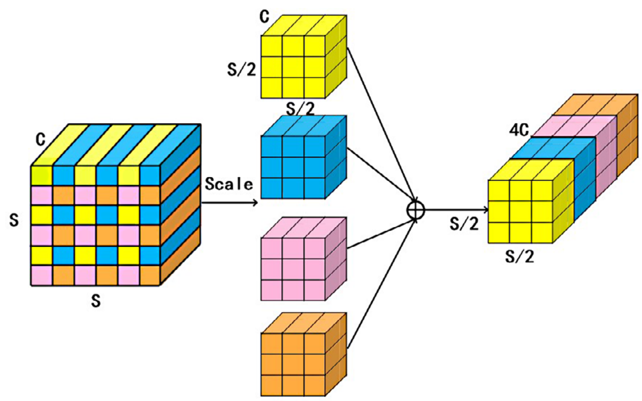

FIGURE 3 SPD module. SPD, space-to-depth.

of information from each pixel, thereby avoiding indiscriminate information loss.

In summary, the SPD-Conv module comprises the SPD layer, responsible for slicing and concatenating operations to downsample the feature map, and a non-stride convolution layer, which further alters the feature map dimension. This combination enables SPD-Conv to effectively mitigate the loss of discriminative information during downsampling of the feature map. With the scale factor of 2 for downsampling, each operation halves the width and height of the image, thereby reducing the feature map's dimensions while preserving key information. This reduces data dimensions effectively while retaining essential features for subsequent layers, enabling the use of a scale factor of 2 to balance processing speed and detection accuracy effectively. Here a scaling factor of 2 (scale = 2) is applied to operate on the feature map. The SPD layer is integrated into the backbone network solely to minimize the loss of fine-grained information and improve the model's accuracy in detecting small objects. The structure of the SPD layer is depicted in Figure 3.

3.3 | WIoUv3

A well-defined bounding box loss function is pivotal in the loss function for object detection task, as it can substantially enhance the detector's performance.

Intersection over Union (IoU) represents the ratio of intersection to union between the predicted box and the ground truth box, quantifying their level of overlap.

$$IoU = \frac{A \cap B}{A \cup B} \tag{1}$$

IoU losses are defined as

$$\mathcal{L}_{IoU} = 1 - IoU \tag{2}$$

However, such IoU loss function exhibits a critical limitation: when there is no overlap between the predicted box and the ground truth box, the IoU becomes 0, leading to gradient disappearance during back propagation. Consequently, numerous studies have proposed enhancements to mitigate these limitations, such as Distance Intersection over Union (DIoU), which introduces a distance metric penalty term, and Complete Intersection over Union (CIoU), which enhances vertical and horizontal consistency on the basis of DIoU, adopted as the bounding box loss function used in the YOLOv8 model. However, CIoU's description of aspect ratio as a relative value lacks clarity, and it fails to consider the balance between the hard and easy samples. In addressing the challenges of object detection, Wise Intersection over Union (WIoU) introduces a dynamic focusing mechanism that integrates an attention mechanism to formulate the bounding box loss. This sophisticated approach effectively mitigates the penalty imposed by geometric metrics when the anchor box closely aligns the object box. This approach results in better generalization ability with minimal intervention during training.

WIoUv1 incorporates distance-based attention mechanisms, resulting in a model with two layers of attention mechanism defined as follows:

$$\mathcal{L}_{WIoU} = \mathcal{R}_{WIoU} \mathcal{L}_{IoU} \tag{3}$$

$$\mathcal{R}_{WIoU} = \exp\left(\frac{(x - x_{gt})^2 + (y - y_{gt})^2}{(W_g^2 + H_g^2)^*}\right) \tag{4}$$

Here, x and y denote the coordinates of the centre point of the detection box, while W_g and H_g denote the size of the minimum bounding box. The superscript * represents separating W_g and H_g from the computed graph, mitigating the penalty term's effect on gradients descent convergence.

Building upon WIoU v1, an additional enhancement was implemented by incorporating a monotonic focal mechanism for cross-entropy. This entailed formulating monotonic focal coefficients to emphasize challenging examples and enhance, thus yielding WIoU v2. In WIoU v3, an outlier factor is introduced and defined as follows:

$$\beta = \frac{\mathcal{L}^*{}_{IoU}}{\mathcal{L}_{IoU}} \in [0, +\infty) \tag{5}$$

When the outlier factor is small, it signifies a high quality of the predicted bounding box. Consequently, a small gradient gain is assigned. Conversely, when dealing with predicted bounding boxes with a large outlier factor, a smaller gradient gain is assigned. This strategy mitigates the risk of low-quality examples producing significant detrimental gradients, thus enabling the bounding box regression to prioritize high-quality predictions. WIoU v3 is defined as follows:

$$\mathcal{L}_{WIoUv3} = r\mathcal{R}_{WIoU} \mathcal{L}_{IoU}, r = \frac{\beta}{\delta \alpha^{\beta - \delta}} \tag{6}$$

In the definition of WIoU v3, α and δ serve as hyperparameters. When β equals δ, r equals *1*. If the outlier factor β of a predicted bounding box matches a specific constant value C, the bounding box is granted the highest gain. This mechanism guarantees the WIoU v3 dynamically adjusts its gradient gain allocation strategy according to the prevailing circumstances.

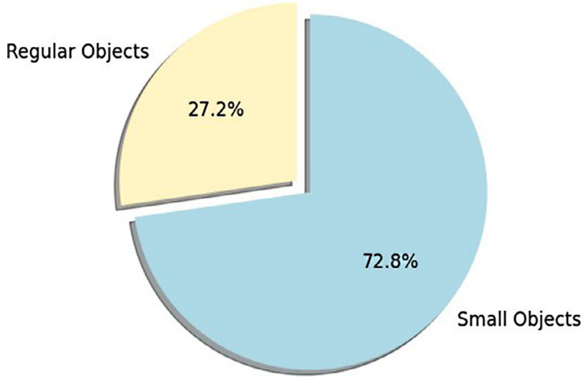

FIGURE 4 The ratio of small object in CCTSDB.

4 | EXPERIMENTS

The experiment utilized the Ubuntu 20.04 operating system, PyTorch 1.12 deep learning framework, and the NVIDIA TITAN Xp Graphics Processing Unit (GPU). Training employed the Stochastic Gradient Descent (SGD) optimizer with an initial learning rate of 0.01, a batch size of 8, and lasted 200 epochs. Furthermore, an automatic mixed precision training strategy was implemented.

The proposed method underwent testing on the CCTSDB and TT100K datasets. CCTSDB comprises over 17,000 images and encompasses three types of traffic signs, including daytime and night-time scenarios. TT100K, another publicly accessible dataset from China, comprises 16,000 images, housing a total of 27,000 traffic sign instances. These datasets facilitate thorough testing and validation of traffic sign detection technologies across diverse conditions.

In this study, instances where the ratio of the object bounding box area to the image area falls within the range of 0.08% to 0.25% are classified as small objects. Statistical analysis was performed on the CCTSDB and TT100K datasets to determine the proportion of these small object samples within the total number of samples. The results are depicted in Figures 4 and 5.

Over 80% of the samples in the CCTSDB and TT100K datasets are classified as small objects. This high occurrence of multi-scale variations and complex background occlusions in the traffic sign data strongly resonates with the research motivation of this study.

4.1 | Evaluation index

To assess the performance of the proposed method against other approaches, we employed the following evaluation metrics.

4.1.1 | Precision and recall

This section outlines the precision and recall metrics, where precision (P) denotes the ration of correctly predicted samples to all predicted samples, and recall (R) indicates the ratio of correctly predicted samples to all positive samples. TP, TN, FP, and FN denote four types of test results.

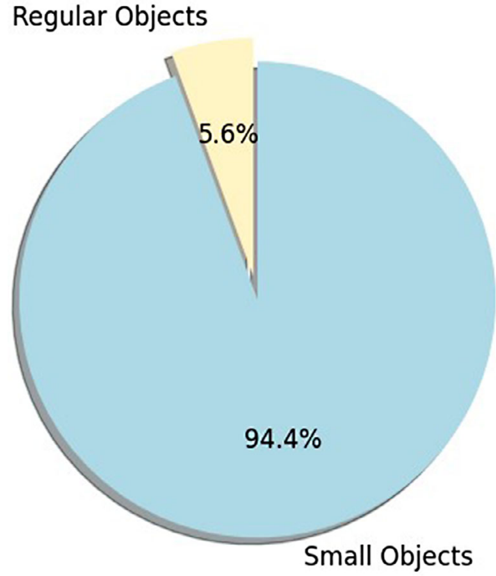

FIGURE 5 The ratio of small object in TT100K.

$$P = \frac{TP}{TP + FP} \tag{7}$$

$$R = \frac{TP}{TP + FN} \tag{8}$$

4.1.2 | mAP

Average Precision (AP) assesses the object detection performance for each class individually by analyzing the precision-recall trade-off. It evaluates the overlap between the predicted and ground truth bounding boxes. The Mean Average Precision (mAP) represents the average AP value across all classes.

During validation and testing, the evaluation metrics typically utilize mAP with an IoU threshold of 0.5 to determine the required level of overlap for a detection to be deemed correct. This assessment encompasses all classes within the dataset.

mAP is calculated by averaging the AP values for each class using an IoU threshold of 0.5. This metric offers a comprehensive evaluation of the detection model's performance, considering both precision and recall. It enables equitable and standardized comparison of different detection models' capacities to accurately detect objects across various classes.

4.1.3 | FLOPs

FLOPs stand for Floating Point Operations, representing the number of floating-point arithmetic operations executed by an algorithm or model, used to gauge its computational complexity. When evaluating a model, we seek a balance among accuracy,

TABLE 1 Ablation experiments on the CCTSDB dataset.

Method	P	R	mAP50	FLOPs(G)	Parameters	FPS
Yolov8s	0.917	0.762	0.849	34.1	7.4 M	**94**
Yolov8s+SPD	0.902	0.781	0.862	47.9	8.4 M	89
Yolov8s+sk	0.925	0.766	0.862	51.7	7.8 M	65
Yolov8s+Wiou	0.919	0.764	0.853	34.1	7.4 M	94
Yolov8s+spd+sk	0.916	0.784	0.863	65.6	8.8 M	62
Yolov8s+spd+Wiou	0.90	0.78	0.862	47.9	7.4 M	89
Yolov8s+sk+Wiou	90.92	50.76	0.863	51.7	7.8 M	65
Yolov8s+spd+sk+Wiou	5**0.918**	7**0.786**	**0.865**	65.6	8.8 M	62

FLOP, floating point operations; FPS, frames per second; mAP, mean average precision; SPD, space-to-depth; YoLo, you only look once.

model size, and computational speed, aiming for a lightweight and efficient model without compromising accuracy. FLOPs serve to quantify the computational demand during a model's forward propagation process.

4.1.4 | FPS

FPS (Frames Per Second) denotes the rate at which image frames are processed per second, equivalent to the number of detection tasks executed by a model within the same time frame. In autonomous driving systems, a minimum of 30 FPS is necessary to meet the real-time performance standard for detection speed. This implies that the model must process and analyze a minimum of 30 frames of input data per second to accommodate the dynamic nature of real-time applications like autonomous driving. Attaining a high FPS is essential to ensure the system's detection capabilities are both prompt and responsive.

4.2 | Ablation experiment

Ablation experiments were conducted to assess the performance enhancement introduced by the proposed method. All experiments utilized an input size of 640 × 640. The ablation experiments conducted in this study compared the performance of YOLOv8s, which was evaluated after a modification that integrated shallow-level features from the P3 layer with the deep-level features. This fusion was directed towards enhancing the detection of small objects. Notably, the ablation experiments omitted the detection head for large objects.

Ablation experiments were performed on the CCTSDB dataset to assess the effects of the SPD module, SK module, and modified WIoU loss function introduced in this paper, as illustrated in the table below

Based on the experimental data presented in Table 1, incorporating the SPD module, SK module, and modified WIoU loss function individually led to an increase in mAP50 values by 1.3%, 1.3%, and 0.4%, respectively. Combining the SK module with the SPD module resulted in a 1.4% improvement in mAP50. Additionally, replacing the WIoU loss function further enhanced model's overall performance, achieving a comprehensive mAP of 86.5% on the CCTSDB dataset, representing a 1.6% improvement compared to the original model.

Figure 6 provides a visual comparison of the mAP50 values among models with the SPD module, SK module, and their combination with WIoUv3, alongside the original model. The figures demonstrate that integrating the SK and SPD resulted in enhanced mAP50 values. Moreover, their combination exhibited a synergistic effect, further boosting the detector's performance. These observations affirm the efficacy of all the introduced modules in this study.

To further illustrate the performance enhancement of each module for detecting small objects, such as traffic signs, this study conducted ablation experiments on the TT100K dataset. The experimental setup remained consistent with previous experiments, and the results are outlined in Table 2. Analysis of the experimental data reveals that integrating the SPD module, SK module, replacing the WIoU loss function, and combining these improvements all contributed to enhancing the overall performance of the original model.

When individually incorporating the SPD module, SK module, and replacing the WIoU loss function, the mAP50 values increased by 1.5%, 1.6%, and 0.4%, respectively. The combination of the SK module with the SPD module yielded a 2.5% improvement in mAP50. Furthermore, by replacing the WIoU loss function, the overall performance of the model was further enhanced, achieving a comprehensive mAP of 90.6% on the TT100K dataset, a 2.7% improvement compared to the original model. Figures 7 illustrates the comparison of mAP50 values between the models with the addition of the SPD module, SK module, and their combination with WIoUv3, and the original model. Similar to the previous dataset, the introduced modules in this study improved the performance of the detector on the TT100K dataset, further validating the effectiveness of the proposed methods.

Based on the data presented in Tables 1 and 2, it is clear that the proposed method exhibits enhanced detection accuracy compared to the original YOLOv8 model. However, there is a reduction in FPS by approximately 30 frames. Nevertheless, the detection speed remains at around 60 frames, which is adequate to meet the real-time demands of autonomous driving scenarios.

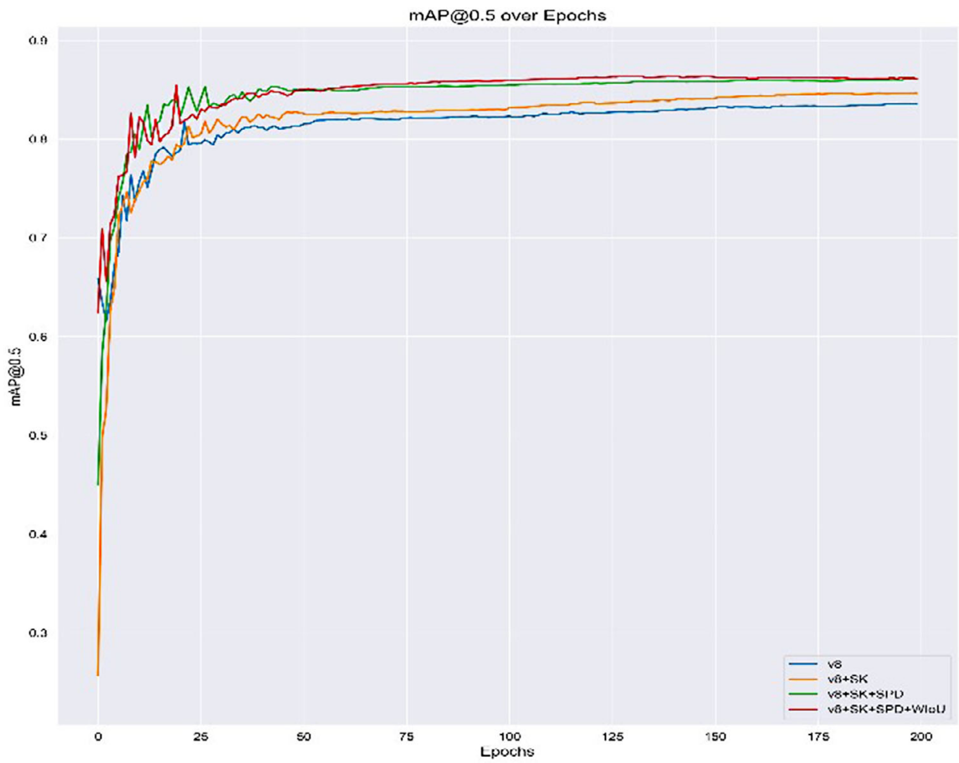

FIGURE 6 Comparison of mAP50 curves. mAP, mean average precision.

TABLE 2 Ablation experiments on the TT100K dataset.

Method	P	R	Map50	FLOPs(G)	Parameters	FPS
Yolov8s	0.868	0.81	0.879	34.2	7.4 M	**93**
Yolov8s+SPD	0.866	50.82	0.893	48.1	8.4 M	76
Yolov8s+sk	0.882	0.82	0.895	51.9	7.8 M	68
Yolov8s+Wiou	0.888	20.79	0.883	34.2	7.4 M	93
Yolov8s+spd+sk	0.877	50.83	0.904	65.7	8.8 M	59
Yolov8s+spd+Wiou	0.893	20.82	0.903	48.1	8.4 M	76
Yolov8s+sk+Wiou	0.89	10.81	0.899	51.9	7.8 M	68
Yolov8s+spd+sk+Wiou	9**0.878**	**0.838**	**0.906**	65.7	8.8 M	59

FLOP, floating point operations; FPS, frames per second; mAP, mean average precision; SPD, space-to-depth; YoLo, you only look once.

The ablation experiments on the CCTSDB and TT100K datasets reveal that the SPD module effectively counteracts the loss of detailed information resulting from the stride convolution operation during feature extraction. By retaining finer features, the SPD module mitigates the decline in detection accuracy attributable to the small size and low resolution of traffic signs in the detection task. This underscores the efficacy of the SPD module in enhancing the detector performance for small objects such as traffic signs.

The experimental results further validate that the WIoU loss function is effective in fitting the regression of bounding boxes, resulting in an improvement in the overall performance of the model. The WIoUv3 loss function enables the model to better handle challenging examples by emphasizing their importance during training. This, in turn, enhances the model's classification performance and contributes to the overall improvement in performance observed in the experiments.

The SK module plays a crucial role in enabling the model to adaptively select the receptive field size that is more suitable for the task at hand. By doing so, the model can focus more on the object region and effectively reduce the interference from external noise and distractions. This adaptability leads to an improvement in the overall detection performance of the model.

Experimental data indicate that although the method proposed in this paper leads to a decrease in FPS, it still meets

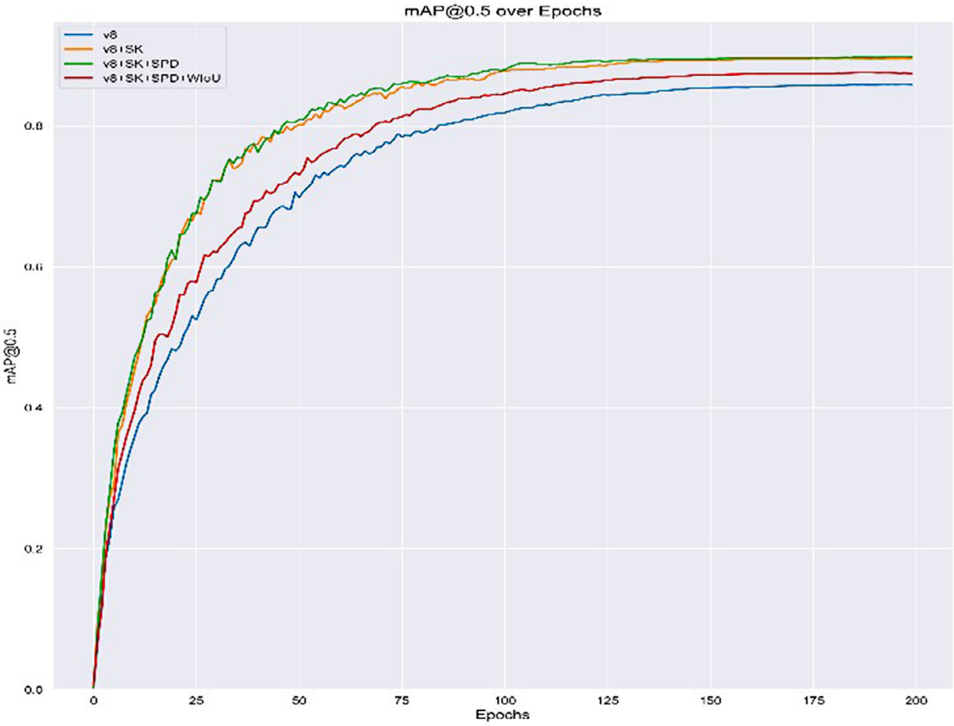

FIGURE 7 Comparison of mAP50 curves. mAP, mean average precision.

the requirements for real-time processing. Compared to this, the improvements in recall and accuracy demonstrate that our method effectively mitigates the issue of missed detections caused by multi-scale transformations and complex background elements, which is crucial for traffic sign detection. Therefore, the approach presented in this paper can effectively enhance the overall detection capability of the model while ensuring that the detection speed meets the demands of real-time detection.

In this study, heatmap visualization was conducted on the feature maps before and after the SK module to illustrate the effectiveness. This visualization technique aids in observing activation patterns and identifying regions of interest in the feature maps. Contrasting the heatmaps before and after the SK module reveals how it improves the model's capability to focus on pertinent regions while suppressing irrelevant information. Such visualization offers qualitative evidence of the SK module's efficacy in enhancing the model's attention mechanism and consequently enhancing the detection performance.

Figures 8 and 9 depict the validation results on the TT100K and CCTSDB datasets, respectively. The figures illustrate that prior to integrating the SK module, the model is influenced by external factors during the detection process, such as roads and vehicles. However, after the data passes through the SK module, the model's attention becomes concentrated on regions containing traffic signs. This highlights the effectiveness of the SK module in filtering out irrelevant information and concentrates on the object areas.

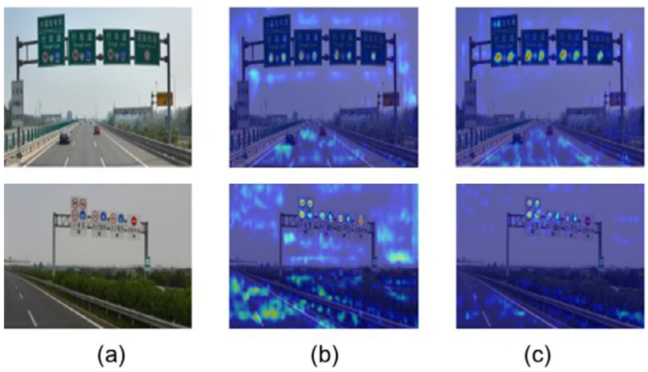

FIGURE 8 Heat map verification of SK module on TT100K datasets. SK, Select Kernel.

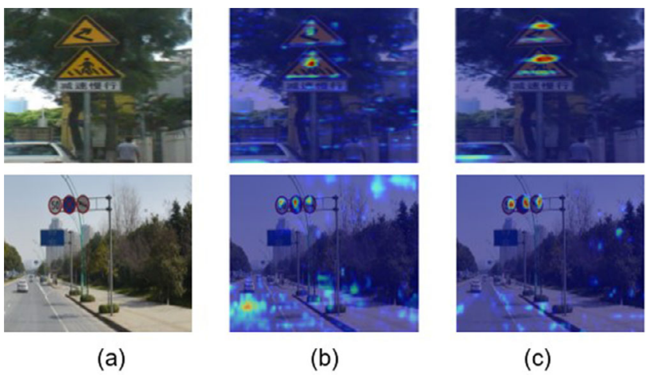

FIGURE 9 Heat map verification of SK module on CCTSDB datasets. SK, Select Kernel.

TABLE 3 Comparison with other models on the CCTSDB dataset.

Method	P	R	mAP50	FPS
yolov8s	0.892	0.748	0.833	**105**
yolov5s	0.910	0.747	0.815	98
yolov7-tiny	0.920	0.723	0.793	102
SSD	0.593	0.515	0.542	28
Faster RCNN	0.769	0.783	0.774	12
R-FCN	0.791	**0.802**	0.797	10
YOLOv7-TS [32]	0.926	0.783	0.860	37
SC-YOLO [33]	**0.938**	0.768	0.843	\
ReYOLO [34]	\	\	0.839	\
Ours	0.918	0.786	**0.865**	62

FPS, frames per second; mAP, mean average precision; YoLo, you only look once.

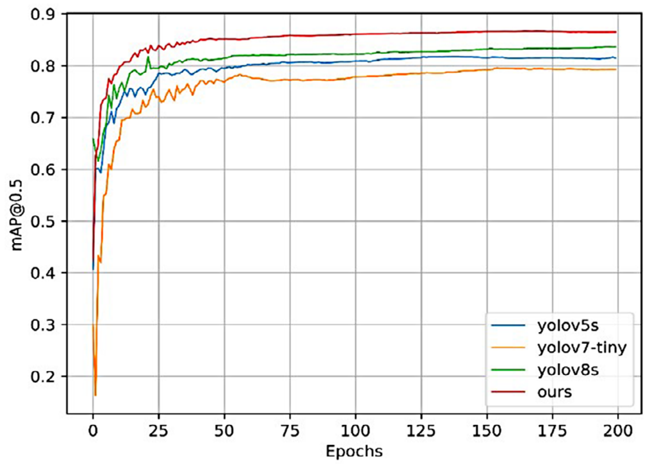

FIGURE 10 Comparison of mAP50 curves with other YOLO models. mAP, mean average precision; YoLo, you only look once.

TABLE 4 Comparison with other models on the TT100K dataset.

Method	P	R	mAP50	FPS
yolov8s	0.863	0.770	0.855	**102**
yolov5s	0.732	0.696	0.741	99
yolov7-tiny	0.530	0.611	0.604	99
SSD	0.548	0.477	0.508	29
Faster RCNN	0.605	0.704	0.698	14
R-FCN	0.638	0.731	0.722	12
Li [28]	\	\	0.901	63
ReYOLO [34]	\	\	0.683	\
Ours	**0.878**	**0.838**	**0.906**	59

FPS, frames per second; mAP, mean average precision; YoLo, you only look once.

4.3 | Comparison with other mainstream models

To validate the performance of the proposed detection algorithm, we compared it with mainstream single-stage detection models. Since the improvements in this study were based on YOLOv8s, a smaller model within the YOLOv8 series, we selected lightweight models for comparison, namely YOLOv5 and YOLOv7 series. Unlike the ablation experiments, we maintained the original detection head of YOLOv8s without any modifications. Each model was tested with an input size of 640 × 640 on both the CCTSDB and TT100K datasets. The experimental results are presented in Tables 3 and 4.

The proposed improvement algorithm, as shown in Table 3, yields substantial accuracy improvements when based on YOLOv8s, compared to mainstream single-stage detection algorithms on the CCTSDB dataset. Notably, in comparison to the original YOLOv8s, the mAP50 value increased by 3.2%. Moreover, compared to YOLOv5s and YOLOv7-tiny, the mAP50 value increased by 5% and 7.2%, respectively. The most significant improvement was observed when compared to SSD, with the mAP50 increasing by 32.3%.

To further validate the robustness of the proposed detection algorithm, we conducted comparative experiments on the TT100K dataset, with the results displayed in Table 4. The proposed model achieves an mAP50 value that surpasses YOLOv8s by 5.1%, YOLOv5s by 16.5%, YOLOv7-tiny by 30.2%, and SSD by 39.8%. Figure 10 illustrates the mAP50 comparison between the proposed method and the latest three versions of the YOLO series on the TT100K dataset, demonstrating the significant improvement in overall performance and faster convergence speed with our approach.

In summary, the proposed enhancement algorithm based on YOLOv8s not only surpasses the original model but also exhibits superior performance compared to other mainstream single-stage detection models.

Figures 11 and 12 respectively display the detection results of the proposed method on the CCTSDB and TT100K datasets. In the figures, column (a) displays the original images, column (b) shows the detection results by YOLOv8, and column (c) presents the detection results of the method proposed in this paper. The illustrations highlight that the proposed method successfully identifies objects missed by YOLOv8's detection, effectively addressing the issue of missed detections and enhancing the detector's accuracy for small objects. Figure 13 compares the miss rates for three object categories and the average miss rate on the CCTSDB dataset. Given the greater variety of object categories in the TT100K dataset, this study selected the eight most numerous categories for a miss rate comparison, the results of which are depicted in Figure 14.

5 | CONCLUSION

In this study, we tackled the challenges posed by multi-scale variations and complex backgrounds in traffic sign detection by making adaptive enhancing YOLOv8s model. By integrating the SPD module into the network backbone, we effectively mitigated the loss of information for small objects during convolution and pooling, thus enhancing the model's ability to

FIGURE 11 Traffic sign detection results on CCTSDB.

FIGURE 12 Traffic sign detection results on TT100K.

detect traffic signs of varying sizes. This integration enabled the model to retain finer details, resulting in enhanced detection performance across diverse objects sizes. Moreover, the SK attention mechanism introduced in this study adjusts the model's receptive field and optimized feature extraction, especially in visually complex scenes. This mechanism allowed the model to preserve crucial features while reducing interference from irrelevant background details, thereby increasing recognition accuracy. Additionally, to address the imbalance in training data distribution, we employed the WIoUv3 loss function. This function dynamically adjusted weights for instances of different sizes and frequencies, accelerating the model's learning for these instances and enhancing overall detection precision.

Experiments conducted on the CCTSDB and TT100K datasets revealed that the implementation of these methods enhanced model performance, notably on the mAP50 metric,

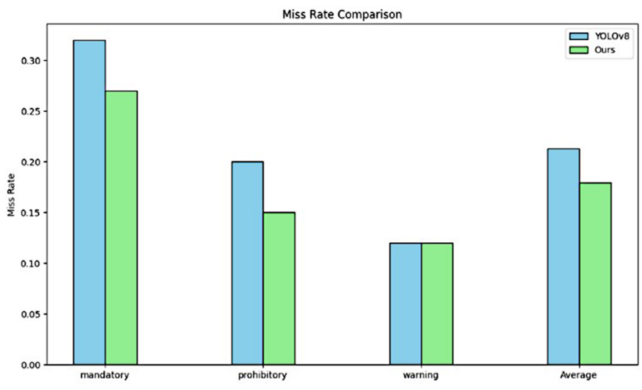

FIGURE 13 Comparison chart of miss rates in CCTSDB.

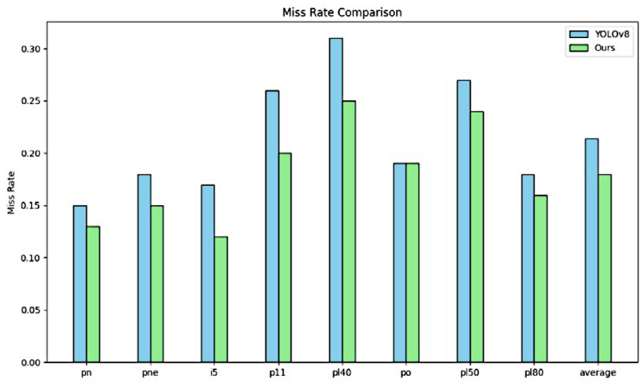

FIGURE 14 Comparison chart of miss rates in TT100K.

with improvements of 3.2% and 5.1% compared to YOLOv8s, respectively. These results validate the effectiveness of the improvements and highlight the model's capability in handling complex traffic scenes.

Future work will prioritize improving the model's robustness under extreme weather conditions and low-light scenarios. Additionally, given the real-time processing demands of autonomous driving systems, efforts will be made to optimize the model's inference speed. It is anticipated that ongoing technological advancements will expand the model's utility across various applications within autonomous driving technology.

AUTHOR CONTRIBUTIONS

Songjie Du: Conceptualization; methodology; software; validation; formal analysis; investigation; data curation; writing—original draft preparation; visualization. **Weiguo Pan**: Conceptualization; formal analysis; resources; writing—review & editing; supervision; project administration; funding acquisition. **Nuoya Li**: Software; data curation. **Songyin Dai**: Resources; supervision. **Bingxin Xu**: Resources; supervision. **Hongzhe Liu**: Resources; supervision; project administration; funding acquisition. **Cheng Xu**: Resources; supervision. **Xuewei Li**: Resources; supervision; project administration; funding acquisition.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

Data available on request.

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